

IsiZulu AI-Detection Classifier — Training & Test Results

Model: fine-tuned binary sequence classifier | Label 1 = machine-written | Label 0 = human-written | Report date: 26 May 2026

1. Training Overview

4

Epochs completed
(early stop at 4 of 5)

2 048

Total steps

0.994

Best eval F1
(checkpoint-1024, epoch 2)

0.9940

Best eval accuracy
(epoch 2)

8

Batch size

2×10^{-5}

Peak learning rate

Epoch-by-epoch validation performance

Epoch	Loss	Accuracy	F1	Precision	Recall	MCC	ROC-AUC	FP	FN
1	0.0951	0.9839	0.9842	0.9698	0.9990	0.9683	0.9997	31	1
2 ★ best	0.0324	0.9940	0.9940	0.9910	0.9970	0.9880	0.9999	9	3
3	0.1868	0.9714	0.9722	0.9459	1.0000	0.9443	0.9997	57	0
4	0.0698	0.9915	0.9915	0.9832	1.0000	0.9831	0.9998	17	0

Early stopping triggered after epoch 4 (patience = 2). Best checkpoint saved at step 1024 (epoch 2).

Final training-subset performance (epoch 2, best checkpoint)

99.80% Train Accuracy	0.9980 Train F1	0.9960 Train Precision
1.0000 Train Recall	0.9960 Train MCC	1.0000 Train ROC-AUC

2. Test Set Results

2.1 Test Set 1 — Held-out balanced set 2 022 samples

Metric	Human (0)	Machine (1)	Overall
Accuracy	—	—	0.9926
F1 Score	0.9926	0.9926	0.9926
Precision	0.9980	0.9873	—
Recall	0.9872	0.9980	—
ROC-AUC	—	—	0.9997
MCC	—	—	0.9852

Confusion Matrix

	Pred: Human	Pred: Machine
Actual: Human	1 000 True Negative	13 False Positive
Actual: Machine	2 False Negative	1 007 True Positive

2.2 Test Set 2 — Large imbalanced set 16 924 samples

Metric	Human (0)	Machine (1)	Overall
Accuracy	—	—	0.9386
F1 Score	0.9681	0.1614	—
Precision	1.0000	0.0878	—
Recall	0.9382	1.0000	—
ROC-AUC	—	—	0.9999
MCC	—	—	0.2870

Confusion Matrix

	Pred: Human	Pred: Machine
Actual: Human	15 785 True Negative	1 039 False Positive
Actual: Machine	0 False Negative	100 True Positive

Note on Test Set 2: This set is heavily imbalanced — only 100 machine-written samples out of 16 924 (0.6%). The model achieves perfect recall for machine text (0 false negatives) but generates 1 039 false positives, inflating the false-positive rate. The high ROC-AUC (0.9999) confirms strong discriminative ability; the low MCC (0.287) and F1-machine (0.161) reflect the impact of class imbalance on headline accuracy metrics. Adjusting the classification threshold would trade recall for precision.

2.3 Test Set 3 — Small balanced set 224 samples

Metric	Human (0)	Machine (1)	Overall
Accuracy	—	—	0.9821
F1 Score	0.9836	0.9804	—
Precision	1.0000	0.9615	—
Recall	0.9677	1.0000	—
ROC-AUC	—	—	1.0000
MCC	—	—	0.9646

Confusion Matrix

	Pred: Human	Pred: Machine
Actual: Human	120 True Negative	4 False Positive
Actual: Machine	0 False Negative	100 True Positive

3. Cross-Set Summary

Dataset	Samples	Accuracy	F1 (macro)	MCC	ROC-AUC	FP	FN	Notes
Validation (best epoch)	~1 990	0.9940	0.9940	0.9880	0.9999	9	3	Strong
Test Set 1	2 022	0.9926	0.9926	0.9852	0.9997	13	2	Strong
Test Set 2	16 924	0.9386	—	0.2870	0.9999	1 039	0	Imbalanced
Test Set 3	224	0.9821	0.9820	0.9646	1.0000	4	0	Strong